

Deploying a Node.js Application Using Nginx as Reverse Proxy

Objective

This project demonstrates a complete multi-server deployment pipeline for a Node.js application using:

- GitHub for source code
- Node.js for application runtime
- Nexus Repository Manager for artifact storage
- Nginx as reverse proxy
- PM2 for process management

Step 1 - Configure Build Server

Objective

The Build Server is responsible for:

- Cloning Node.js application
- Installing dependencies
- Building/package application
- Uploading artifact to Nexus

1.1 Update Server

Command:

```
sudo apt update -y
```

```
sudo apt upgrade -y
```

1.2 Install Node.js

Install Node.js 20:

1.3 Install Git

Command:

```
sudo apt install git -y
```

1.4 Clone Node.js Application

```
git clone <NODEJS_GITHUB_REPO_URL>
```

1.5 Install Dependencies

Command:

```
npm install
```

1.6 Build Application

Command:

```
npm run build
```

```
Node.js v22.22.1
ubuntu@BULID:~/AngularCalculator$ export NODE_OPTIONS=--openssl-legacy-provider
-bash: export: '--openssl': not a valid identifier
-bash: export: '--legacy': not a valid identifier
-bash: export: '--provider': not a valid identifier
-bash: export: '--openssl-legacy-provider': not a valid identifier
ubuntu@BULID:~/AngularCalculator$ export NODE_OPTIONS=--openssl-legacy-provider
ubuntu@BULID:~/AngularCalculator$ ng build
node: --openssl-legacy-provider is not allowed in NODE_OPTIONS
ubuntu@BULID:~/AngularCalculator$ export NODE_OPTIONS=--openssl-legacy-provider
ubuntu@BULID:~/AngularCalculator$ export NODE_OPTIONS=--openssl-legacy-provider
ubuntu@BULID:~/AngularCalculator$ unset NODE_OPTIONS
ubuntu@BULID:~/AngularCalculator$ export NODE_OPTIONS=--openssl-legacy-provider
ubuntu@BULID:~/AngularCalculator$ echo $NODE_OPTIONS
--openssl-legacy-provider
ubuntu@BULID:~/AngularCalculator$ ng build
18% building 4/4 modules 0 active(node:12588) [DEP0111] DeprecationWarning: Access to process.binding('http_parser') is deprecated.
(Use 'node --trace-deprecation ...' to show where the warning was created)
(node:12588) [DEP0111] DeprecationWarning: Access to process.binding('http_parser') is deprecated.
(node:12588) [DEP0060] DeprecationWarning: The 'util._extend' API is deprecated. Please use Object.assign() instead.

Date: 2026-05-18T14:13:58.029Z
Hash: df285638c9c085e24033
Time: 1147ms
chunk [es2015-polyfills] es2015-polyfills.js, es2015-polyfills.js.map (es2015-polyfills) 285 kB [initial] [rendered]
chunk [main] main.js, main.js.map (main) 10.7 kB [initial] [rendered]
chunk [polyfills] polyfills.js, polyfills.js.map (polyfills) 236 kB [initial] [rendered]
chunk [runtime] runtime.js, runtime.js.map (runtime) 6.88 kB [entry] [rendered]
chunk [styles] styles.js, styles.js.map (styles) 971 kB [initial] [rendered]
chunk [vendor] vendor.js, vendor.js.map (vendor) 3.2 MB [initial] [rendered]
ubuntu@BULID:~/AngularCalculator$ Connection to ec2-3-148-286-48.us-east-2.compute.amazonaws.com closed by remote host.
Connection to ec2-3-148-286-48.us-east-2.compute.amazonaws.com closed.
PS C:\Users\User\Downloads> ssh -i "my.pem" ubuntu@ec2-3-148-286-48.us-east-2.compute.amazonaws.com
```

1.7 Package Application

Create artefact.

Command:

```
tar -czvf nodeapp.tar.gz
```

Step 2 - Configure Nexus Repository Server

Objective

Store Node.js artifacts centrally.

2.1 Install Java

Command:

```
sudo apt update -y
```

```
sudo apt install openjdk-17-jdk -y
```

2.2 Download Nexus

Command:

```
wget https://download.sonatype.com/nexus/3/latest-unix.tar.gz
```

Extract: `tar -xvf latest-unix.tar.gz`

2.3 Start Nexus

Command:

```
sudo /opt/nexus/bin/nexus start
```

Check:

```
sudo /opt/nexus/bin/nexus status
```

2.4 Access Nexus UI

Browser: `http://<NEXUS_SERVER_IP>:8081`

Default Username: admin

Get Password: `cat /opt/sonatype-work/nexus3/admin.password`

2.5 Create Hosted Repository

Navigate:

Settings → Repositories → Create Repository

Create:

- raw-hosted

Why raw-hosted?

- Node.js artifacts like `.tar.gz` are stored as raw files.

Step 3 - Upload Artifact to Nexus

Objective

Upload packaged Node.js application to Nexus.

3.1 Upload Using curl

From Build Server:

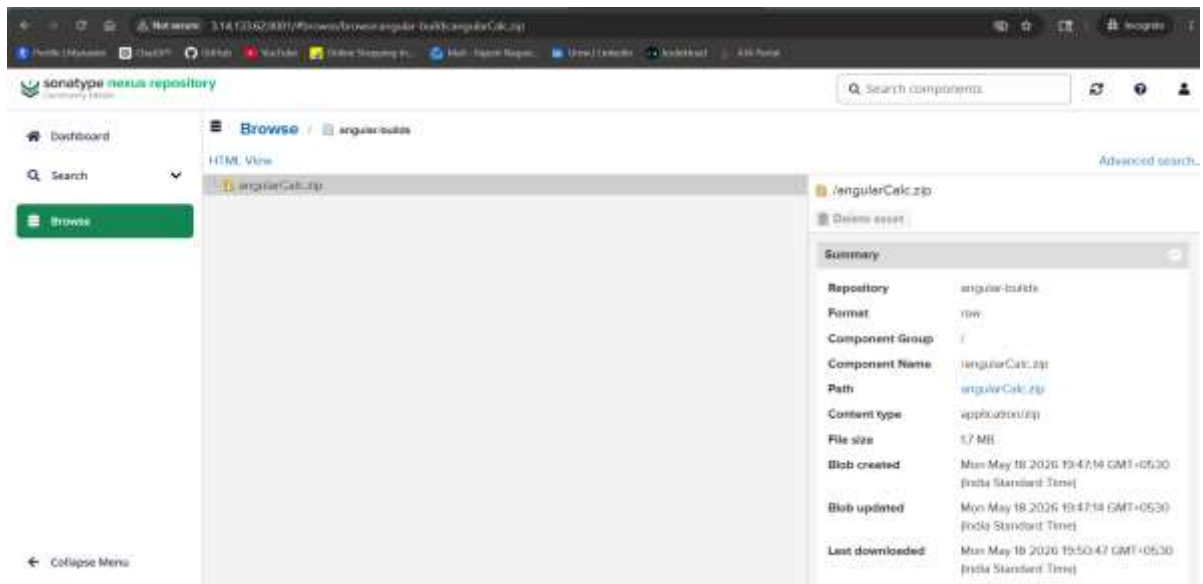
```
curl -v -u admin:admin123 --upload-file nodeapp.tar.gz \
http://<NEXUS_SERVER_IP>:8081/repository/raw-hosted/nodeapp.tar.gz
```

```
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@nexus-repo:~$ unzip angularCalc.zip
Archive: angularCalc.zip
  creating: angularCalc/
  inflating: angularCalc/polyfills.js
  inflating: angularCalc/index.html
  inflating: angularCalc/runtime.js.map
  inflating: angularCalc/styles.js
  inflating: angularCalc/styles.js.map
  inflating: angularCalc/main.js.map
  inflating: angularCalc/main.js
  inflating: angularCalc/polyfills.js.map
  inflating: angularCalc/vendor.js
  inflating: angularCalc/es2015-polyfills.js
  inflating: angularCalc/runtime.js
  inflating: angularCalc/es2015-polyfills.js.map
  inflating: angularCalc/vendor.js.map
  inflating: angularCalc/favicon.ico
ubuntu@nexus-repo:~$ ls
angularCalc  angularCalc.zip  nexus-3.85.0-03  nexus-3.85.0-03-linux-x86_64.tar.gz  sonatype-work
ubuntu@nexus-repo:~$ ls angularCalc
es2015-polyfills.js  favicon.ico  main.js  polyfills.js  runtime.js  styles.js  vendor.js
es2015-polyfills.js.map  index.html  main.js.map  polyfills.js.map  runtime.js.map  styles.js.map  vendor.js.map
ubuntu@nexus-repo:~$ sudo apt install nodejs npm -y
Installing:
 nodejs  npm
Installing dependencies:
```

3.2 Verify in Nexus

Navigate:

Browse → raw-hosted



Step 4 - Configure Deploy Server

Objective

Deploy and run Node.js application using:

- Nginx Reverse Proxy

4.1 Install Node.js

Command:

```
sudo apt install nodejs -y
```

4.2 Install PM2

PM2 keeps Node.js app running continuously.

Command:

```
sudo npm install -g pm2
```

4.3 Install Nginx

Command:

```
sudo apt install nginx -y
```

```
ubuntu@deploy:~$ ls
angularCalc  angularCalc.zip  apache-tomcat-9.0.118  apache-tomcat-9.0.118.tar.gz
ubuntu@deploy:~$ sudo apt install nginx -y
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap  nginx-doc  ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 28
  Download size: 662 kB
  Space needed: 1974 kB / 4025 MB available

Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 nginx-common all 1.28.3-2ubuntu1.1 [16.5 kB]
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu/resolute-updates/main amd64 nginx amd64 1.28.3-2ubuntu1.1 [625 kB]
Fetched 662 kB in 8s (9465 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 98639 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.28.3-2ubuntu1.1_all.deb ...
Unpacking nginx-common (1.28.3-2ubuntu1.1) ...
```

Command:

```
sudo systemctl start nginx
sudo systemctl enable nginx
```

```
Last login: Mon May 18 15:18:23 2026 from 104.28.155.34
ubuntu@deploy:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-05-19 05:04:47 UTC; 17min ago
   Invocation: 2cfff6fdd89e54b378e5d3ce2c1c326de
     Docs: man:nginx(8)
   Process: 606 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 633 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Main PID: 670 (nginx)
    Tasks: 3 (limit: 627)
   Memory: 4.5M (peak: 4.7M)
      CPU: 42ms
   CGroup: /system.slice/nginx.service
           └─670 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─672 "nginx: worker process"
               └─673 "nginx: worker process"

May 19 05:04:47 deploy systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
May 19 05:04:47 deploy systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@deploy:~$
```

Step 5 - Download Artifact from Nexus

5.1 Download Artifact

Command:

```
wget http://< 3.137.199.19>:80/repository/raw-hosted/nodeapp.tar.gz
```

5.2 Extract Artifact

Command:

```
tar -xvf nodeapp.tar.gz
```

```
wget http://admin:admin123@3.14.133.62:8081/repository/angular-builds/angularCalc.zip
ls
unzip
sudo apt install unzip
unzip angularCalc.zip
```

5.3 Restart Nginx

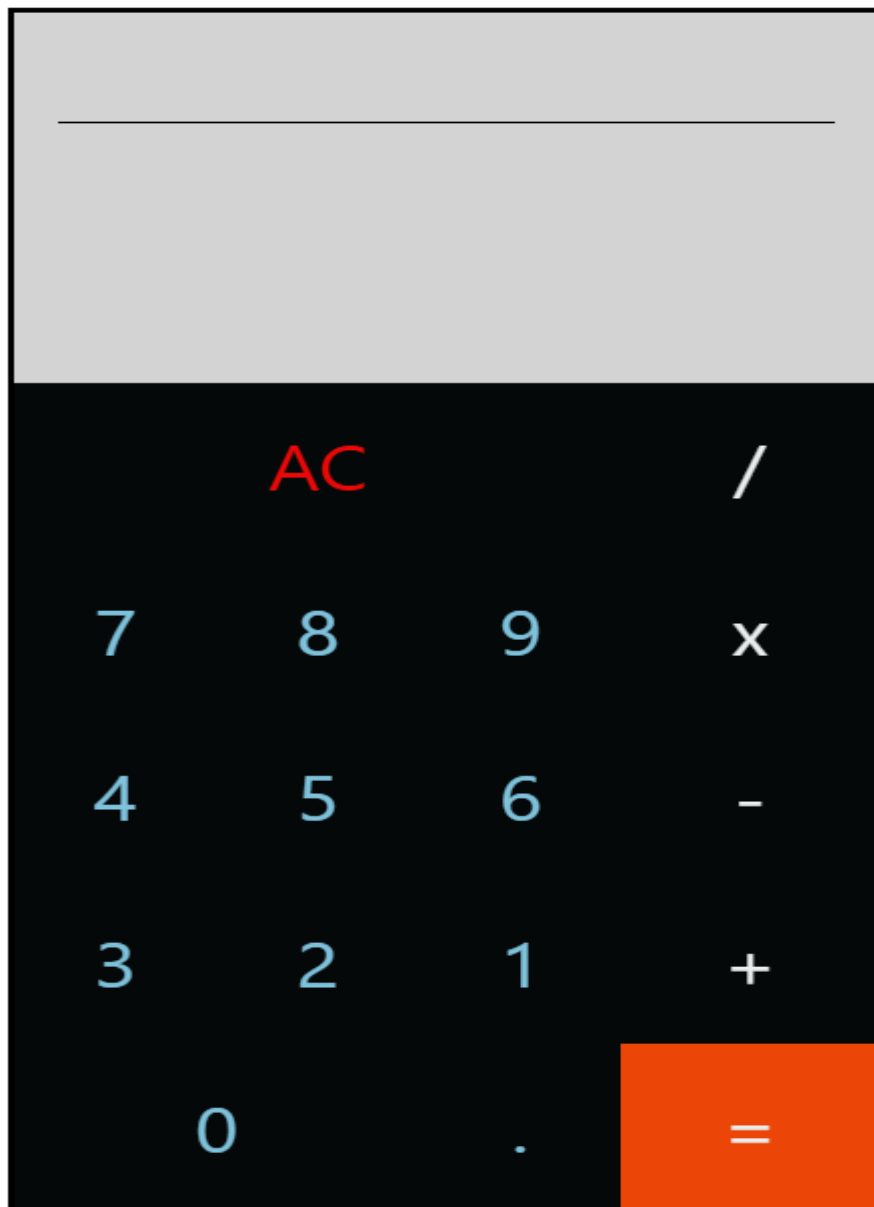
Command:

```
sudo systemctl restart nginx
```

Step 6- Access Application

Browser: <http://< 3.137.199.19>80>

Nginx will forward traffic to Node.js app running on port 80



Conclusion

This project successfully implemented a production-style Node.js deployment pipeline using:

- GitHub
- Node.js
- Nexus Repository
- PM2
- Nginx Reverse Proxy

The architecture provides:

- Centralized artifact storage
- Automated deployments
- Scalable reverse proxy setup
- Better application availability and traffic management